



Public nEUro

A European repository to share neuroimaging datasets publicly.

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novo
nordisk
fonden
Benefiting people and society

Septembre 9-10 2026

We are a Data Repository

[PublicnEUro.eu](#) is not a research environment (ie it does not compute), it provides an option for researchers to share their data*. (for brain imaging data, only when obtained for research purposes).

Host and publishes (DOI) BIDS datasets, making them findable and accessible.

Public Data: making a dataset information public (open metadata, controlled access for download).

Options range from fully open to fully closed (embargo period for consortia, but title and description can be open) – basically almost everything is optional (because we are a service)

We are a Data Processor

(yes there is a small fee for service and hosting)

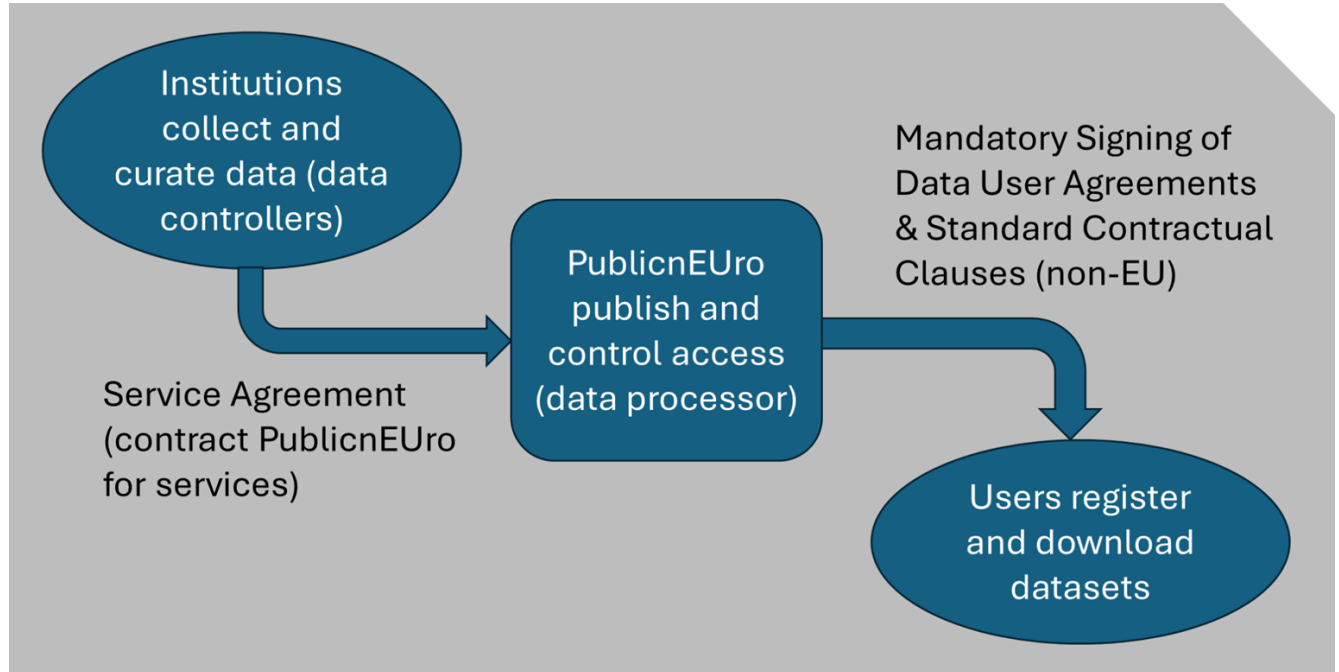


Figure 1: overview of the data workflow (boxes and arrows) and legal tools around it

FAIR data sharing

Finding data within and across repositories

Repository findability: Metadata and metadata curation

Access data in practice: Upload and download

Accessibility policies

Finding data within and across repositories

Finding data within the repository

[Content](#) [Datasets \(24\)](#) [Funding](#)

↓ Name

↓ Modified

Filter by dataset or author name

Filter by keyword (select or press enter)

Keywords

 [PN000001 OpenNeuroPET Phantoms](#)
Søren Baarsgaard Hansen, Murat Bilgel, Keith Ciantar, Anthony Galassi, Gabriel Gonzalez-Escamilla, Sune Høggild Kelle, Maqsood Yaqub, Cyril Pernet
Version: V2 **Modified:** 2025-11-14

Positron Emission Tomography PET Brain
Source files ecat7 DICOM Phantom
Conversion PET2BIDS dcm2niix

 [PN000002 Aggression Project](#)
Sofia da Cunha-Bang, Liv Vadskjaer Hjort, Erik Perfall, Camilla Bock, Dorte M. Sestoft, Patrick Fisher, Gitte Moos Knudsen
Version: V1 **Modified:** 2024-09-15

functional MRI HCP-Young Adults
multiverse analytical choices motor task

 [PN000003 HCP multipipelines](#)
Elodie Germani, Elisa Fromont, Pierre Maurel, Camille Maumet
Version: V1 **Modified:** 2024-10-02

functional MRI HCP-Young Adults
multiverse FSL motor task

 [PN000004 Closed Loop Sleep Motor Memory](#)
Judith Nicolas, Gaëlle Leroux
Version: V1 **Modified:** 2025-02-15

sleep motor learning

- Browse datasets
- Search by authors
- Search by keywords

Finding data within the repository



Aggression Project

Sofi da Cunha-Bang, Liv Vadskjaer Hjordt, Erik Perfalk, Camilla Bock, Dorte M. Sestoft, Patrick Fisher, Gitte Moos Knudsen

Version: V1 **Last updated:** 2024-09-15 **DOI:** [10.70883/DEYZ5967](https://doi.org/10.70883/DEYZ5967) **License:** Data User Agreement **Status:** Active

Cite

Export metadata

Share

Download

Description: The Aggression Project aimed to determine brain mechanisms underlying aggression using 5-HT1BR PET ([11C]AZ10419369) and MRI. Participants recruited included incarcerated violent offenders and healthy controls. (total size: 16.0GB)

Keywords:

functional MRI

HCP-Young Adults

multiverse

analytical choices

motor task

Content

Datasets (0)

Publications

Funding

Dataset Metadata

Participants

DUA terms

dataset_description.json	1.07 KB
participants.tsv	2.13 KB
participants.json	1.14 KB
task-faces_events.json	2.22 KB
task-psap_events.json	1.11 KB
task-rest_events.json	2 Bytes

sub-53273

sub-53273_scans.tsv	549 Bytes
---------------------	-----------

pet

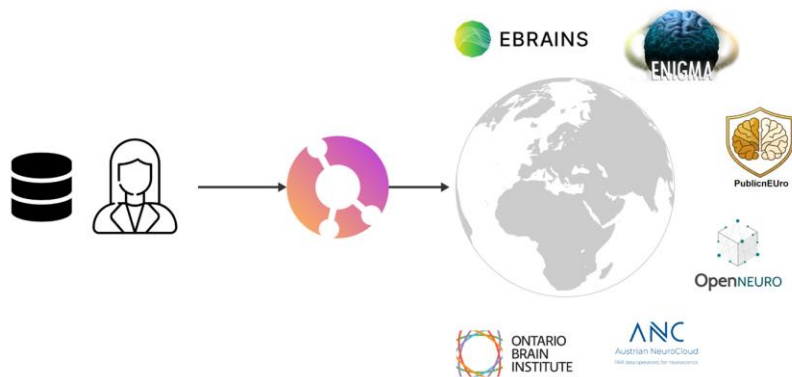
fmap

anat

func

- Browsing file representations
- Metadata and Participant summary information

Finding data across repositories



Neurobagel node

-- need to push datasets

BIDS2EBRAINS

-- need to push datasets
and fix issues if any

```
{
  "@context": {"@vocab":
    "https://openminds.ebrains.eu/vocab/"},
  "@id":
    "https://kg.ebrains.eu/api/instances/<uuid>",
  "@type":
    "https://openminds.ebrains.eu/core/FileRepository"
,
  "IRI": "file://local-placeholder",
  "hostedBy": {"@id":
    "https://kg.ebrains.eu/instances/organizationalUnit/
    ebrains"},
  "name": "local-placeholder"
}
```


Repository findability: Metadata and metadata curation

Concretely



Closed Loop Sleep Motor Memory

Judith Nicolas, Gaëlle Leroux

Version: V1 **Last updated:** 2025-02-15 **DOI:** [10.70883/QXBH2876](https://doi.org/10.70883/QXBH2876) **License:** Data User Agreement **Status:** Active

[Cite](#) [Export metadata](#) [Share](#)

[Download](#)

Description: Unraveling the Neurophysiological Correlates of Phase-Specific Enhancement of Motor Memory Consolidation via Slow-Wave Closed-Loop Targeted Memory Reactivation (total size: 81.0GB)

Keywords:

[sleep](#) [motor learning](#)

[Content](#) [Datasets \(0\)](#) [Publications](#) [Funding](#) [Dataset Metadata](#) [Participants](#) [DUA terms](#)

dataset_description.json	2.21 KB
task-SRTTtesting_events.json	3.39 KB
task-SRTTtraining_events.json	3.39 KB
participants.tsv	534 Bytes
participants.json	480 Bytes
phenotype	
extra_data	
sub-01	
sub-02	
sub-03	

- Each dataset has a description
- PublicnEUro schema also have a **status**
- We provide a DOI
- Datasets come with licenses or DUA → PublicnEUro translates those in **Digital Use Conditions**

<https://github.com/Public-nEUro/PublicnEUro-metadata>



Closed Loop Sleep Motor Memory

Judith Nicolas, Gaëlle Leroux

Version: V1 **Last updated:** *unknown* **DOI:** [10.70883/QXBH2876](https://doi.org/10.70883/QXBH2876) **License:** Data User Agreement

” Cite

Export metadata

Share

Download

Description: Unraveling the Neurophysiological Correlates of Phase-Specific Enhancement of Motor Memory Consolidation via Slow-Wave Closed-Loop Targeted Memory Reactivation

Keywords:

sleep

motor learning

Content

Datasets (0)

Publications

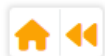
Funding

Dataset Metadata

Participants

DUA terms

Property	Value(s)
Bids_version	v1.9.0
Bids_datasettype	raw
Bids_datatypes	anat func eeg
NCBI Species Taxonomy	homo sapiens
Cognitive Atlas concept	motor sequence learning
Cognitive Atlas task	serial reaction time task rest eyes open Pittsburgh Sleep Quality Index



Closed Loop Sleep Motor Memory

Judith Nicolas, Gaëlle Leroux

Version: V1 **Last updated:** *unknown* **DOI:** [10.70883/QXBH2876](https://doi.org/10.70883/QXBH2876) **License:** Data User Agreement

Cite

Export metadata

Share

Download

Description: Unraveling the Neurophysiological Correlates of Phase-Specific Enhancement of Motor Memory Consolidation via Slow-Wave Closed-Loop Targeted Memory Reactivation

Keywords:

sleep

motor learning

Content

Datasets (0)

Publications

Funding

Dataset Metadata

Participants

DUA terms

Property	Value(s)
Total_number	31
Age_range	[18, 30]
Number_of_healthy	31
Number_of_biological_males	16
Number_of_biological_females	15

PublicNEuro metadata

Machine-readable, dataset-level governance metadata derived from the [PublicNEuro DataCatalogue](#).

The generated records keep three independent facts for every dataset version:

- status:** active, archived, retired, withdrawn, Or superseded;
- retrieval mode:** online, cold_archive, external, Or unavailable;
- digital use conditions (DUC):** the licence or Data User Agreement summary exposed by the catalogue.

For backwards compatibility, a catalogue record without **status** is exported as **active**.

24 datasets / 25 versions

- Access:** 3 open access / 21 restricted
- Participants:** 2,130 total (1,731 healthy / 399 patients)
- Documented size:** 5.06 TB (reported for 23/24 datasets)

Figures use the latest version of each dataset. Access is restricted when the catalogue licence is **Data User Agreement**; participant counts come from the **Participants** panel; patients are inferred as total minus healthy participants. Counts are dataset-level and may include the same individuals in related datasets.

Dataset	Version	Status	Retrieval	DUC	DOI	Updated
PN000001 OpenNeuroPET Phantoms	V1	superseded	online	1 reviewed condition	10.70883/IGQP1334	2024-06-02
PN000001 OpenNeuroPET Phantoms	V2	active	online	1 reviewed condition	10.70883/RGNV3768	2025-11-14
PN000002 Aggression Project	V1	active	online	9 reviewed conditions	10.70883/DEYZ5967	2024-09-15
PN000003 HCP multipipelines	V1	active	online	7 reviewed conditions	10.70883/GTKK1541	2024-10-02
PN000004 Closed Loop Sleep Motor Memory	V1	active	online	6 reviewed conditions	10.70883/QX8H2876	2025-02-15

```
> curation
v datasets
  PN000001.json
  PN000002.json
  PN000003.json
  PN000004.json
  PN000005.json
  PN000006.json
  PN000007.json
  PN000008.json
  PN000009.json
  PN000010.json
```

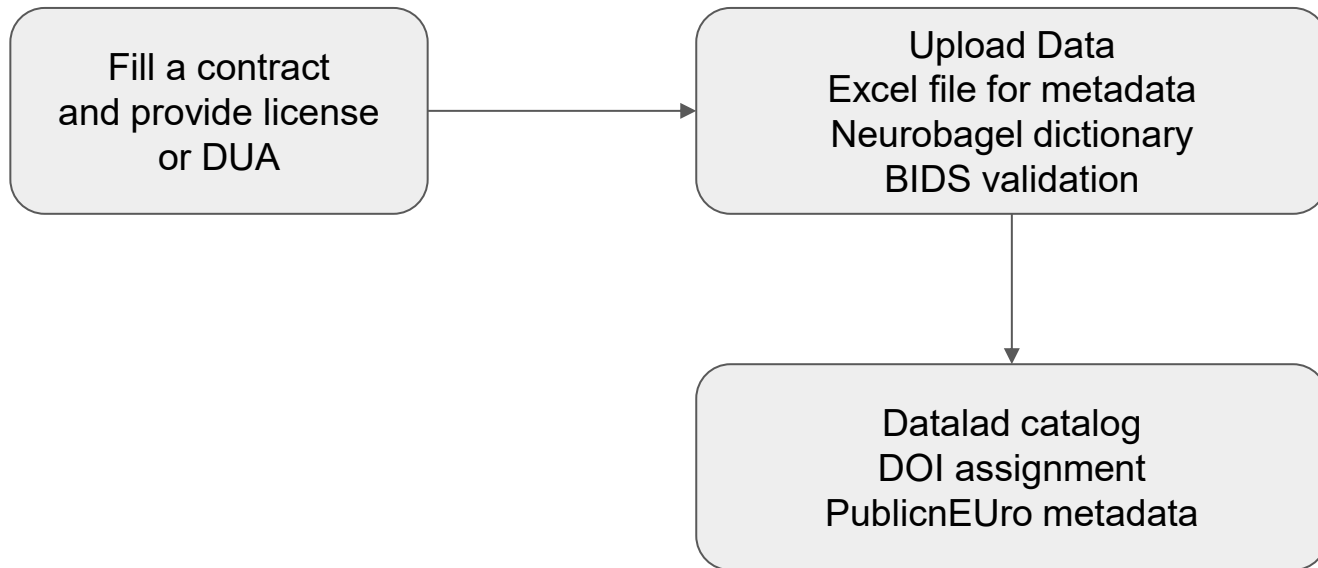
```
1 {
2   "schemaVersion": "1.3",
3   "datasetId": "PN000002",
4   "name": "Aggression Project",
5   "versions": [
6     {
7       "version": "V1",
8       "catalogueUrl": "https://datacatalog.publicneuro.eu/dataset/PN000002",
9       "status": "active",
10      "retrieval": {
11        "mode": "online",
12        "url": "https://datacatalog.publicneuro.eu/manage/request-a",
13      },
14      "duc": {
15        "profileVersion": "0.1.0",
16        "profileName": "PN000002 V1 Digital Use Conditions",
17        "ducVersion": "1.1.0",
18        "permissionMode": "All unstated conditions are Permitted",
19        "language": "eng",
20        "assets": [
```

```
> curation
> datasets
v exports
  openaire-cerif.xml
  re3data.xml
  repository-summary.json
v mappings
  README.md
  dua_to_duc.json
> schema
> scripts
> tests
  .gitignore
```

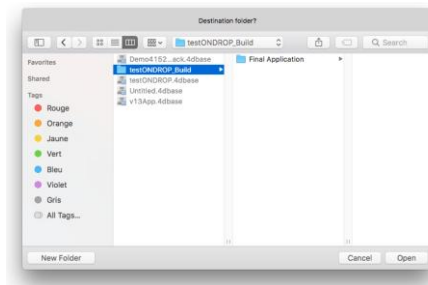
```
1 {
2   "schemaVersion": "1.0",
3   "datasets": 24,
4   "versions": 25,
5   "access": {
6     "open": 3,
7     "restricted": 21,
8     "unclassified": 0
9   },
10  "participants": {
11    "total": 2130,
12    "healthy": 1731,
13    "patients": 399,
14    "datasetsWithCounts": 23
15  },
16  "documentedSize": {
17    "terabytes": 5.06,
18    "datasetsWithSize": 23
19  }
20 }
```

Access data in practice: Upload, Download and After

Upload is done only via GUI / drag and drop



Select a folder and wait



but time out can happen 😞
Too large = many uploads

2 options to access data



Or CLI (always work better but users .. you know)

```
bash <(wget -qO-  
https://datacatalog.publicneuro.eu/api/cli.sh)
```

This will prompt you to enter

your email

your password

the PublicnEUro (PN) ID

the folder that you want to download - use / for the entire dataset

But tar.gz built – too large means split download ☹

Data in the life cycle

- People contract us and pay to share their data for N years
- What's next → keep it hot, archive on tape, take it back

- Mapped onto our data status

```
• status: active , archived , retired , withdrawn , or superseded ;  
• retrieval mode: online , cold_archive , external , or unavailable ;
```

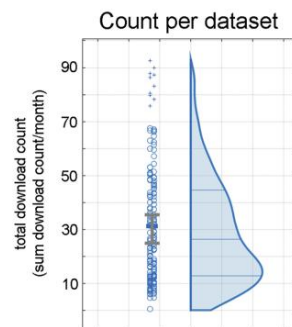
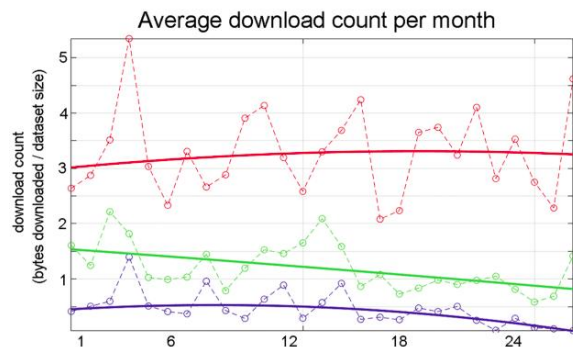
Data access is tracked → does it worth to keep data just active but also hot?

Neuroinformatics (2023) 21:243–246
<https://doi.org/10.1007/s12021-023-09621-x>

EDITORIAL

On the Long-term Archiving of Research Data

Cyril Pernet¹ · Claus Svarer¹ · Ross Blair² · John D. Van Horn³ · Russell A. Poldrack²

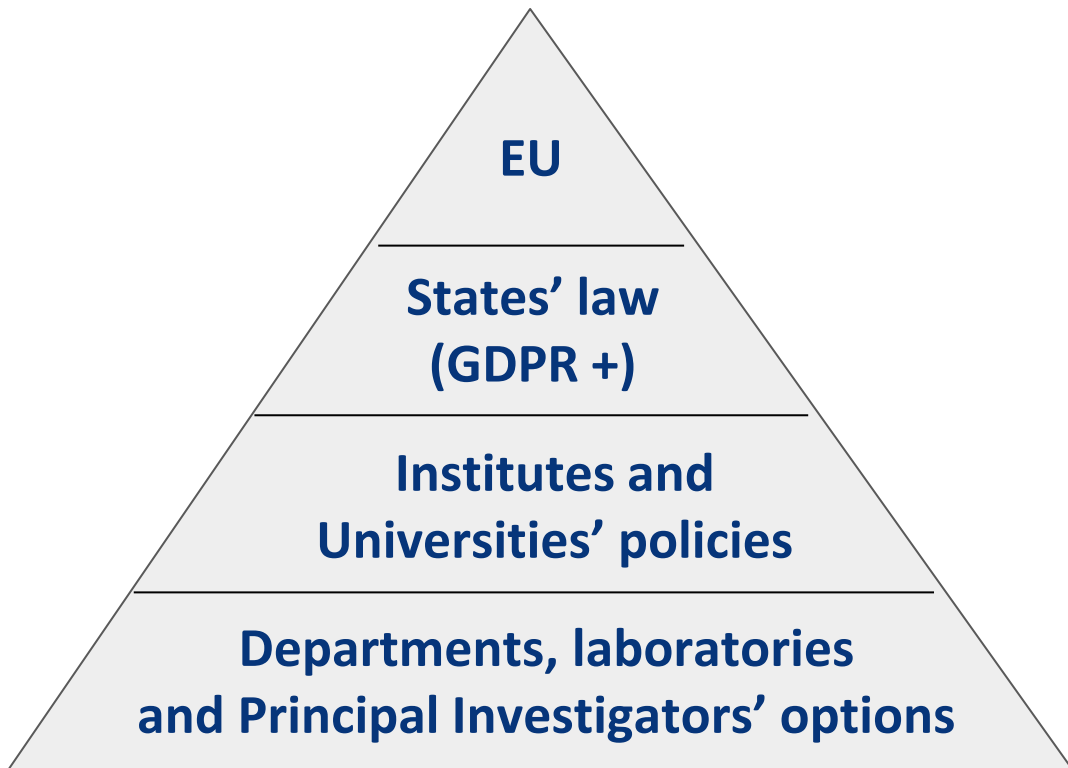


Accessibility policies

GDPR, states, institutions and labs

GDPR declares that within EU personal data can be shared (Art 1), it also states that general conditions do not apply to research data (Art 5 and recital 50), i.e. scientific data can be shared more broadly without purpose limitation (despite layers reluctance).

YOU CAN SHARE - but each state, institution, lab has their own implementation trickling down to you!



GDPR, states, institutions and labs

Each state has a specific implementation adding different restrictions. Each institution has specific policies and data Protection Officers interpret the law and those policies differently. Each lab has specific views on how they want to control data.

By controlling the data users geolocation and by letting data controllers set their 'terms and conditions', PublicnEUro accommodates the different regulations.

Restrictions	Give data processor rights (e.g. with oversight)	Give data controller rights (e.g. without oversight)
EU / EU+	Most restrictive	
World wide		Most permissive

Doc signing

SSC are institutional and exchanged by email and stored - Institutions are then marked as accepting SSC

Request access to dataset

Name	PN000002 Aggression Project		
Accessibility	WORLDWIDE		
DUA	DUA_NRU_for controllers.docx	Signed DUA	Choose File No file chosen

You need to upload the signed DUA before requesting access

[Request access](#)

DUA can be signed online or uploaded (data controller validated)

Request access to dataset

Name	PN000004 Closed Loop Sleep Motor Memory		
Accessibility	WORLDWIDE		
DUA	DUA_ClosedLoop_Sleep_MotorMemory.txt	☒ I accept the DUA	

[Request access](#)

